PSPNet

권세은

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CONTENTS

[1] Zhao, H. et al. "Pyramid scene parsing network." *Computer Vision and Pattern Recognition*. 2017.

INTRODUCTION

▪ Scene Parsing Task

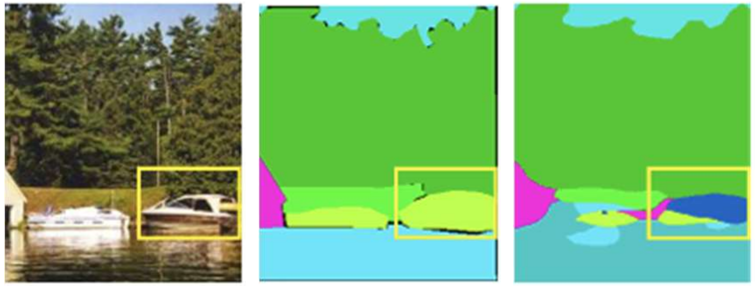
- Semantically segment, interpret scene images
- More classes and unrestricted open vocabulary

  e.g. PASCAL VOC: 21 classes vs ADE20K: 150 classes

  ⇒ more challenging

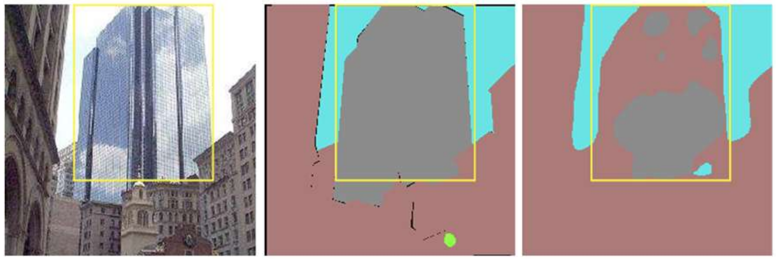
- **Main Problems**

1. Mismatched Relationship



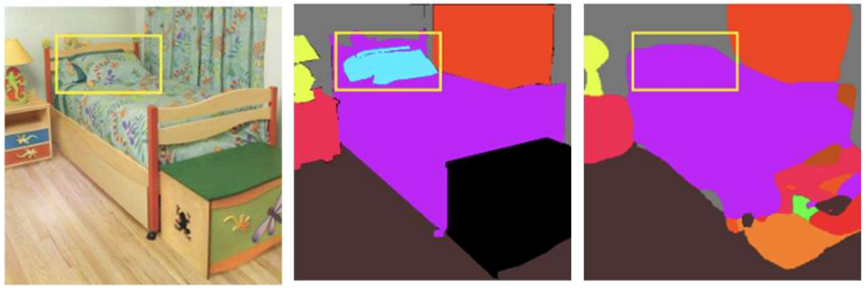
→ boat to car mismatching

2. Confusing similar categories



→ skyscrapper and building confusion

3. Nonrecognition of inconspicuous Classes



→ missing pillow

How to solve?

- context relationship
- categories relationship
- more attention to different sub-regions

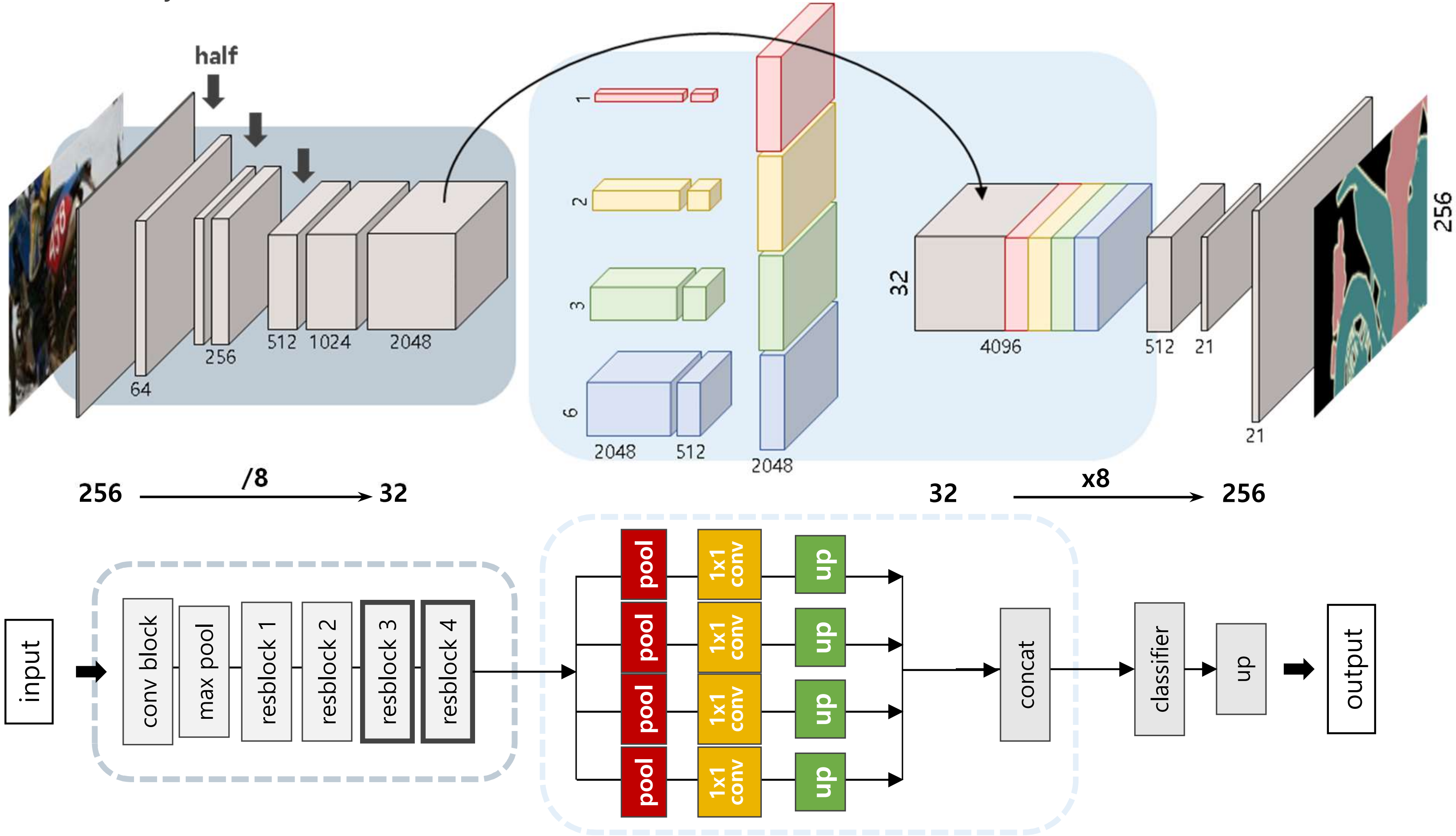
AIVS 인공지능 및 비전 시스템

- **Pyramid Scene Parsing Network**

  ResNet + Pyramid Pooling Module

  - dilated convolution → larger receptive field
  - multi-scale pooling → different sub-region information

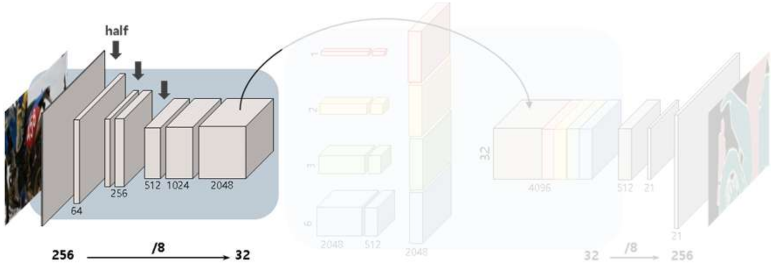
  ⇒ effectively extract **contextual information**

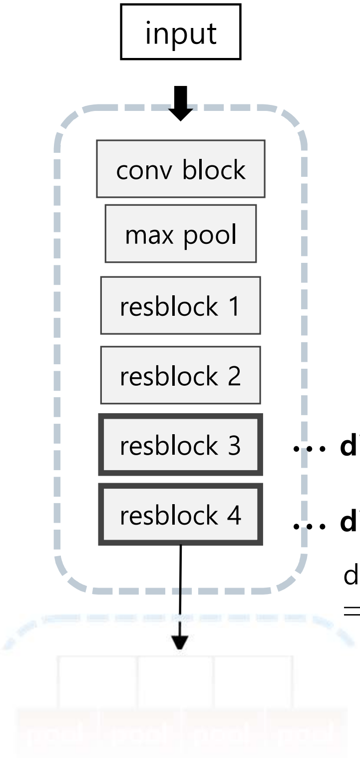


- **Pyramid Scene Parsing Network**

ResNet as a backbone feature extractor

- Dilated convolution for larger receptive field
- Pretrained weight to enhance training efficiency



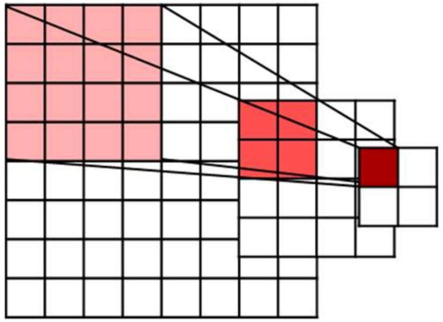


… **dilation=2**

… **dilation=4**

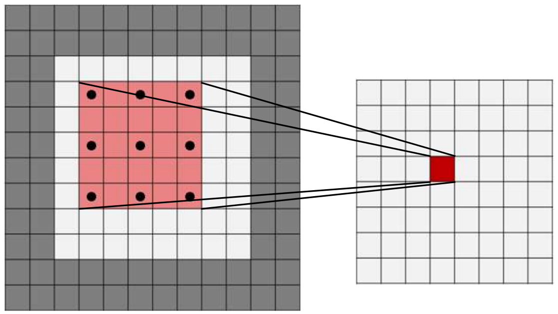
different dilation rate
⇒ different receptive field size

Conv(stride=2)&Pool



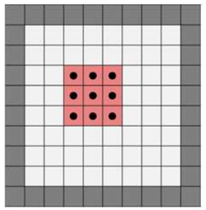
Feature map size↓
⬇
Receptive field size↑
⬇
Include context information

Dilated Conv



dilation=padding=2

* Original Conv



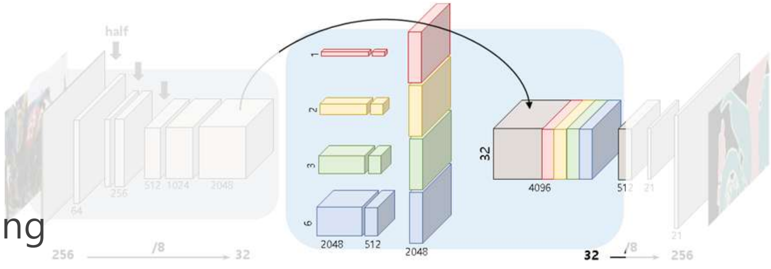
dilation=padding=1

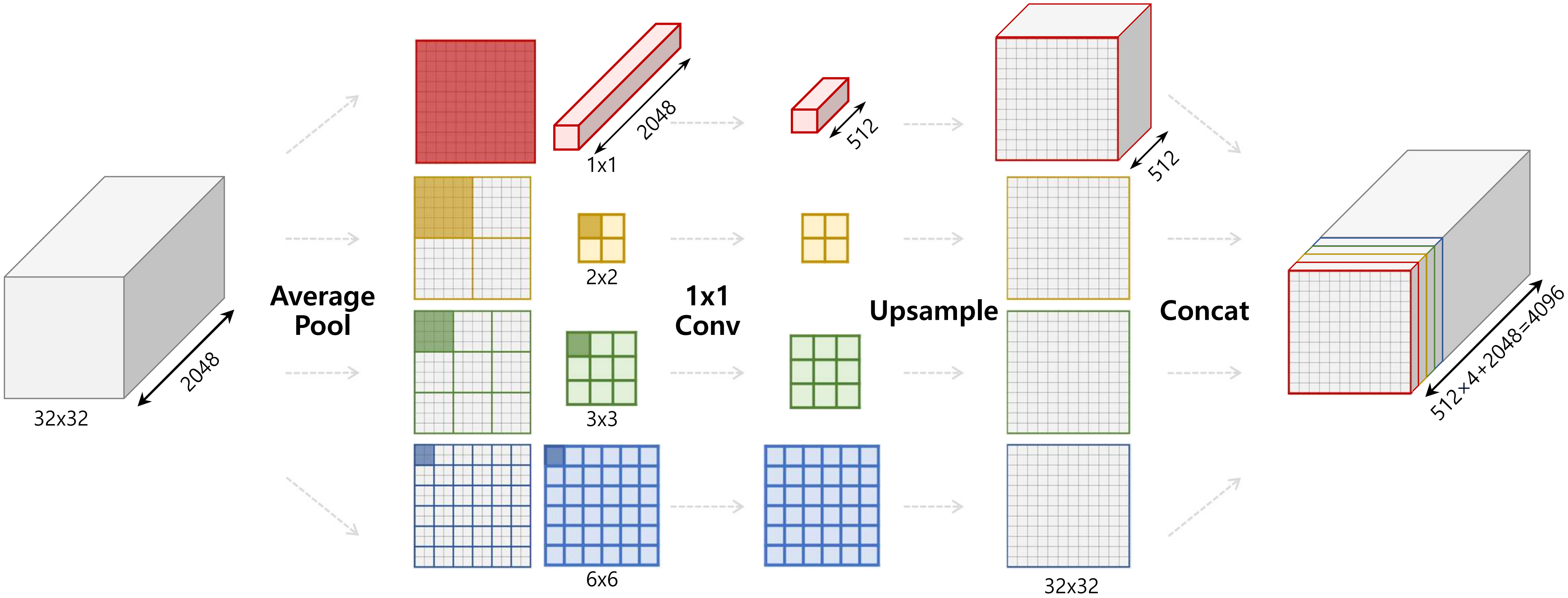
Feature map size maintained ⬇ • Spatial information maintained • Sufficient size for PPM input	Dilated filter ⬇ Receptive filed size↑ ⬇ Include context information

- **Pyramid Scene Parsing Network**

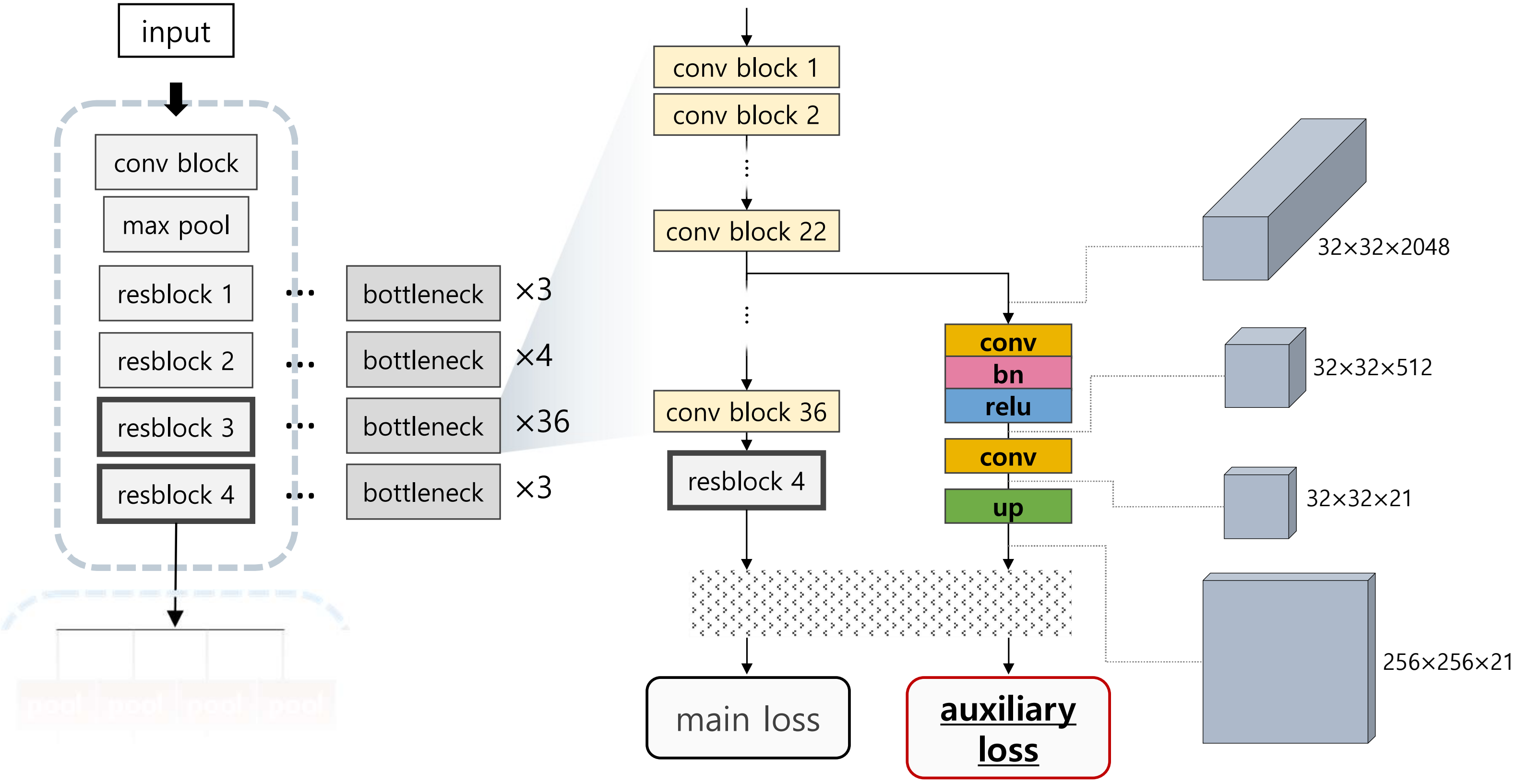
Pyramid Pooling Module

1. **Multi-scale pooling** to capture hierarchical context
2. **1x1 Convolution** to reduce dimension and prevent overfitting
3. **Upsampling** to match original features
4. **Concatenation** to merge multi-scale and original features





- **Auxiliary Loss**
  - $\mathrm{Loss} = \mathrm{L_{main}} + \alpha \cdot \mathrm{L_{aux}}$ (α=0.4)
  - Help optimizing at deep ResNet (layer=101 or 152)



AIVS 인공지능 및 비전 시스템

▪ Setup

	PSPNet
Dataset	PASCAL VOC2012
Iteration	30K
Batchsize	32
Learning rate	$lr_{init}\left(1-\frac{\text{iter}}{\text{max}_{\text{iter}}}\right)^{power}$ $(\boldsymbol{lr_{init}} = \mathbf{0.01}, power = 0.9)$
Optimizer	SGD
Criterion	Cross Entropy Loss
Augmentation	• HSV adjustment • Horizontal/ Vertical flip • Gaussianblur • Rotation(-10~10˚)
Hardware	NVIDIA GeForce RTX 4090

EXPERIMENT

- **Result**

	PA	mIoU
ResNet50	90.69	59.78
ResNet152	91.13	61.12
ResNet152+AL	91.47	62.44

	background	aeroplane	bicycle	bird	boat	bottle	bus	car	cat	chair	cow
Res50	90.07	76.71	**33.80**	71.76	58.37	64.46	78.71	71.63	80.63	**17.02**	57.17
Res152	90.37	76.40	**31.65**	71.74	54.21	67.12	80.12	76.20	81.97	**16.95**	62.52
Res152 +AL	90.98	77.94	**35.12**	78.24	61.86	71.69	82.86	75.88	81.24	**16.86**	56.55

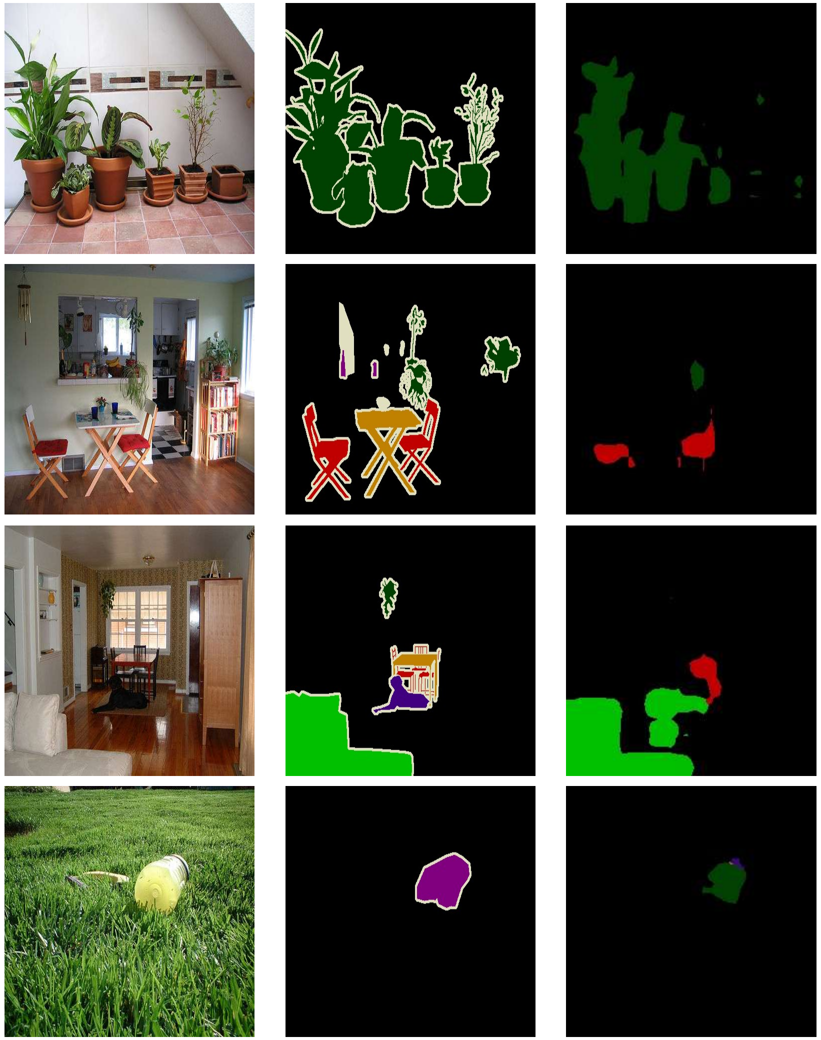
	dinigtable	dog	horse	motorbike	person	potted plant	sheep	sofa	train	tv/ monitor	mIoU
Res50	**35.55**	70.38	59.51	64.03	68.18	**26.22**	67.79	**36.49**	68.00	58.96	**59.78**
Res152	**36.40**	74.14	61.71	61.98	70.79	**31.28**	66.19	**38.72**	70.09	63.08	**61.12**
Res152 +AL	**39.01**	69.68	59.12	66.30	72.10	**33.78**	66.41	**41.36**	70.38	63.97	**62.44**

EXPERIMENT

- **Result**

  ### Output of ResNet152+AL

Input image | Ground Truth | Prediction



Input image | Ground Truth | Prediction

small , thin objects were not recognized well

⇒ more attention needed
to inconspicuous classes

Conclusion

- **Future work**

Adversarial Attack

- Ian J. Goodfellow et al. “EXPLAINING AND HARNESSINGADVERSARIAL EXAMPLES.” *ICLR*, 2015.

- J. H. Metzen et al. ”Universal Adversarial Perturbations Against Semantic Image Segmentation.” ICCV, 2017.

Thank you